Musanze Harvest Intelligence Service

SWE 3513 — Individual Assignment 2 | Technical Report

Student: Shyaka Peter Guy
Registration: 24/26559
Institutional username: ug2527965@ines.ac.rw
Verification code: AI-A2-6559

Decision problem

The prototype supports a potato cooperative by estimating harvest quantity, identifying a high-risk label, and grouping farms with similar feature conditions. It is a classroom decision-support prototype and is not an agronomic authority.

Ethical boundary

The data are entirely synthetic. This report does not claim that measured scores represent performance on real farms. Predictions must not determine access to finance, insurance, public services, or farmer eligibility; a qualified human must review any operational interpretation.

2. Data and reproducibility

The generator uses NumPy's default_rng with the last four registration digits as its seed. The row count is 240 + (6559 mod 41) before deliberate duplicates.

Verification code: **AI-A2-6559**. Input rows: 283; clean rows after exact duplicate removal: 280; exact duplicate rows detected: 3.

Missing values before cleaning: rainfall_mm: 0, avg_temp_c: 0, soil_moisture_pct: 14, fertilizer_kg: 15, field_size_ha: 0, season: 0, cooperative_support: 0, yield_tons: 0, high_risk: 0

SHA-256 of generated CSV: `8c80583da571ebf24b9ff9daded408cbbb0c898a706137d8b2c15472a332f12e`

Features include rainfall, average temperature, soil moisture, fertilizer, field size, season and cooperative support. The target yield follows the assignment formula, including season effect and Gaussian noise, clipped at 0.50 tons. High risk is assigned when yield per hectare is below its median.

The generator deliberately inserts missing values into soil moisture and fertilizer and appends three exact duplicate rows. The pipeline records these problems and removes exact duplicates. Numeric missing values are imputed inside a training pipeline; categorical season is imputed and one-hot encoded.

3. Pipeline design

The run_all.py entry point accepts --student-id and --seed. It rejects a seed other than 6559, regenerates the CSV, validates and cleans the data, fits the models, saves artifacts and prints the personal verification code and fingerprint.

The cleaned dataset is split into training and held-out test partitions using random_state=6559 and stratification for the binary label. Imputers, scalers and encoders are part of scikit-learn pipelines and are fitted only on training data for supervised models.

Target leakage is controlled by using only the seven declared input features. yield_tons and high_risk are excluded from classifier features; target columns are also excluded from clustering. Saved pipelines are loaded unchanged at inference time.

The generator's exact formula is: 0.004×rainfall + 0.06×soil moisture + 0.02×fertilizer + 1.30×field size − 0.25×|temperature−18| + 0.70×support + season effect + noise. Season A contributes +0.50 and B −0.20; noise has mean 0 and standard deviation 0.80.

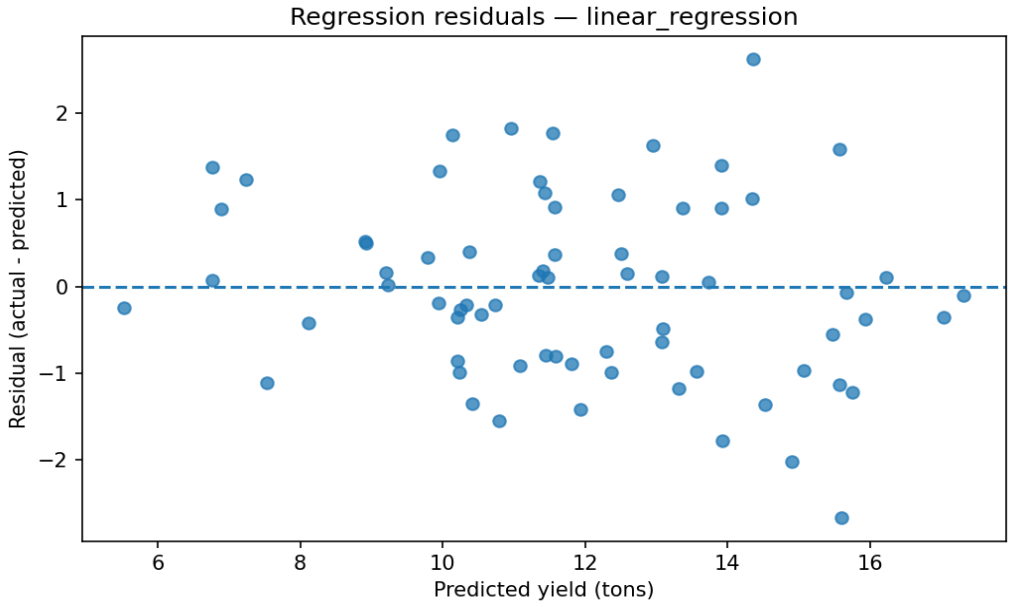
4. Regression and classification

Regressor	MAE	RMSE	R^2
mean_baseline	2.150	2.654	-0.005
linear_regression	0.838	1.040	0.846
random_forest	1.088	1.341	0.743

Selected regressor: **linear_regression**, chosen by the lowest held-out MAE among the two candidates. MAE is the mean absolute error in tons; RMSE penalizes larger errors more strongly; R^2 describes variance explained on this held-out synthetic sample.

Classifier	Precision	Recall	F1	ROC-AUC
majority_baseline	0.000	0.000	0.000	0.500
logistic_regression	0.912	0.886	0.899	0.976
random_forest	0.971	0.943	0.957	0.986

Selected classifier: **random_forest**, chosen by the highest held-out F1 among the candidate classifiers. Precision measures the fraction of predicted high-risk cases that are positive; recall measures the fraction of positive cases found. The confusion matrix makes false positives and false negatives visible.



5. Clustering

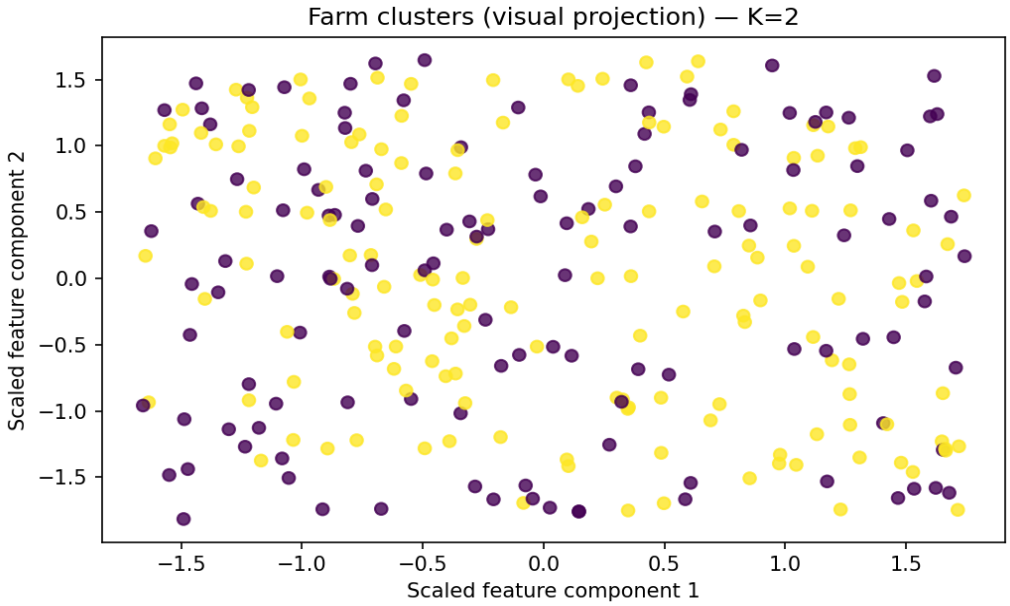
Silhouette scores on scaled input features:

K	Silhouette
2	0.1627
3	0.1406
4	0.1521
5	0.1544
6	0.1560

Selected K = 2, the K with the largest silhouette score among 2–6. Clustering uses feature inputs only; the yield and risk targets are not used to form clusters.

Cluster 0 (124 farms): mean rainfall 976.22 mm; mean moisture 53.16%; mean fertilizer 117.23 kg; mean field size 2.02 ha; common season A. Relatively wetter farms.

Cluster 1 (156 farms): mean rainfall 999.58 mm; mean moisture 52.46%; mean fertilizer 121.63 kg; mean field size 2.35 ha; common season A. Relatively drier farms.



6. API and testing

GET /health returns service status. **GET /model-info** returns the verification code, selected models and synthetic-data notice. **POST /predict** accepts seven typed features and returns estimated yield, high-risk label, probability, cluster and verification code.

Pydantic validates field types and ranges. Unknown fields are forbidden. Invalid requests return a 4xx response (normally HTTP 422). The local CLI uses the same saved pipelines and performs explicit record validation.

The automated tests cover valid inference, missing fields, wrong types, range errors, unexpected fields, health endpoint, predict endpoint, API range failure, and deterministic data generation/seed rejection. The test log in evidence/ records the actual execution output from the build environment; rerun pytest -q in the target environment before submission.

Example valid values: rainfall 950 mm, temperature 18.5°C, moisture 55%, fertilizer 120 kg, field size 1.4 ha, season A, cooperative support 1. Example invalid request: rainfall 1500 mm, which exceeds the allowed 600–1400 range.

7. Model card and reflection

Intended use: classroom demonstration of a reproducible regression, classification, clustering and API workflow.

Data status: synthetic formula-generated records; not representative evidence of Rwandan farms.

Limitations and risks: generated relationships are simplified and may not reflect agronomy, weather variation, measurement error, pests, disease, local practices or economic context. Small test sets can produce unstable metrics. Clusters are descriptive groupings, not farm quality rankings.

Prohibited use: do not use the output to decide credit, insurance, subsidy, public service access, farm eligibility or other consequential decisions. Do not present estimates as guaranteed yields.

Human review: an agronomist or trained field officer must review inputs and interpret predictions. A real deployment requires representative, consented data, external validation, subgroup error analysis, security/privacy review, monitoring and a documented appeal/correction process.

Reflection: the most important next step is to replace synthetic data with responsibly collected, quality-controlled farm data and evaluate against a time- and location-separated holdout set. Error examples should be discussed using artifacts/test_predictions.csv rather than selecting anecdotes that exaggerate performance.

Submission evidence note: the required continuous narrated demo video and final Git commit hash must be created by the student after locally running and committing this project. They are not invented in this package.